

Lec 33 GAN Objective and Loss Functions

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Lecture	33 / 63 + intro
Duration	43:38
Captions	English (auto-generated)
Words	6,407
Video	GqbgTaJcl38
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone, welcome to the lecture series on fundamentals of generative AI and large language models. We are in the week five. In the previous session, we had started with the GAN architecture and then we had seen in detail how generator works and how discriminator works. In this particular uh session, we will be looking into the GAN objective and then the loss function. Although some of the concepts will be repetitive but you will have a complete clarity of the entire working mechanism of GAN mathematically in this particular session. Therefore in this session we'll be seeing what is the discriminator objective function what is generator uh generator objective function and together what is the GAN objective

[00:44] function. So let's get started. So if you look into this particular uh uh slide here what we are looking is the gener uh discriminator uh the loss function. Now when you look into the discriminator discriminator has two parts one where we will be passing the real samples. So when it is a real sample look at this the training data set contains the real samples which are drawn from this particular distribution that is from an unrow unknown real data distribution. So X is sampled from this particular distribution. What is the meaning of X is sampled? We are taking some samples from

this distribution from the real data set and then what is that we are doing? we are passing the uh data to a discriminator. Here if you see X

[01:36] represents one single sample from the training data set. So that X is passed to an discriminator. Here you can see from the diagram that X is passed to an discriminator. Right? So when X is passed to a discriminator what is that discriminator has to do? You are passing an real samples. Right? Therefore the discriminator is going to produce this which is nothing but what what exactly is $d_{\theta}(x)$? It is a predicted probability for the real sample x . So therefore $d_{\theta}(x)$ is a predicted probability that the input x is real.

[02:18] Now just recall from our previous session if x is a real sample what should be its ground truth? The ground truth should be y equal to one. Right? So therefore since x is a real training sample its target label y is one. So because y is one. Now immediately recall now we need to have an binary cross entropy function. So we have this is the binary cross entropy. Now $\hat{y}$ is the predicted one. So this is $d_{\theta}(x)$ which is $\hat{y}$ and y here is one. Can we substitute all of these values into this formula? So substituting $y = 1$ and $\hat{y}$ is $d_{\theta}(x)$. On substituting we are going to get this particular. So the entire since y is 1 this term becomes zero. So when this term becomes zero 0 multiplied with anything becomes zero.

[03:16] So we don't have the second term at all. Therefore we are going to get for a real sample the loss function what we are going to get is this minus $\log d_{\theta}(x)$ this we have seen in the previous session as well but this is required for today's discussion also. So first we will be seeing a quick uh recap. Yeah. So this is all about the discriminator loss for a real sample. So this is the discriminator loss. Now let us see what will happen when we pass the um you know uh samples to an generator. Now how do you pass the samples? What kind of samples we'll be passing? The very first one is you remember first we take the z z is an noise vector here. So that z from where do we take z is sampled

[04:10] from this particular distribution noise distribution. So we can also call this as an normal distribution standard distribution. Right? So first z is sampled from this distribution where z now becomes one random noise vector. Now this z to what you will be passing z is now passed to an generator. Right? The sampled noise vector z is a passed to an generator. This is generator. Now once uh z is passed to two and a generator what is that the generator will give us? Generator will give us the generated data or generated sample. So the generator produces the generated sample. This is what is the generated sample. We can even call it as an $\hat{x}$ or x_{fake} . Okay. This is what we have obtained. So we have now reached to this

[05:03] step. Once we get the generated sample, what is the next step? we have to pass it to an discriminator, right? The generated sample is then passed to an discriminator. So when generated samples are passed to a discriminator, what is that

discriminator has to produce? The discriminator has to give us this, right? What exactly is this? This is the predicted probability for the generated samples. So the gener discriminator produces this. And what exactly is this particular term? This is an predicted probability for the generated samples.

[05:41] Once again recall if it is an generated sample we got y_{hat} . Now you have to compare it with the ground truth. What should be the ground truth? If it is an generated sample y should be equivalent to zero. Right? So look at this. So since we got the generated since this is a generated sample the target for um the target label y is zero. Now once again go back and then take a binary cross entropy. This is binary cross entropy function. For this you take y as zero.

[06:16] And what is y_{hat} ? y_{hat} is this entire the predicted value. So substituting $y = 0$ and y_{hat} this value into an binary cross entropy we are going to get this right. So since $y = 0$ the entire term is going to be zero here the first term is going to be zero. So what is that we are left out we are left out with this particular. So this is what is the loss function for a discriminator when the fake samples are given. So this is what we have uh seen. Now what is the uh next step? So uh once we get the loss functions for real samples and generated samples the total loss L_D is uh the addition of L_{real} and L_{fake} or $L_{\text{generated}}$ samples. So this is what is the total loss L_D equal to L_{real} plus L_{fake} . Now we have already seen from our

[07:21] previous two slides that L_{real} is given by this formula and L_{fake} is this particular formula. Now what is that we are doing? Just take these two formulas and then substitute for L_{real} and L_{fake} we are going to get this and since there is a negative we are taking negative term common. By taking negative term common we are going to obtain this. So can we say that this is the loss function for an discriminator where for a discriminator you are passing real samples as well as generated samples. So this is the total loss of a discriminator. Now uh you if you recall from the basics always we this generator and discriminator here are a neural networks. So what is the aim of a neural network? Ultimately the

[08:11] aim here is to minimize the loss function. So how is that what what loss you have to minimize here? Here you have to minimize the discriminator loss. How is that you will minimize the discriminator loss? You have to update the parameters. What parameters? Aim here is to update the discriminator uh uh improve the discriminator which means discriminator parameters have to be updated. What are discriminator parameters? θ . Now by updating discriminator parameters is it not we are aiming to reduce the uh loss function here. So this is what we are talking about neural networks are usually trained to minimize the loss function and the discriminator parameters are θ have been updated so that the loss becomes as small as

[08:56] possible. So which means your aim is to minimize the loss. So can we write the same L_D now can we write in this uh form? Look at this. It looks very complicated

but it's very very simple. So what exactly is our aim? Our aim is to minimize the loss. So uh here minimum minimum minimize with respect to what? With respect to the parameters and here parameters are denoted as θ . What is that you have to minimize? You have to minimize the loss function. Which loss function? LD that is a discriminator loss function. Now what exactly are these? There are two parameters. One is θ . Second is ϕ . θ is with respect to discriminator. ϕ is with respect to generator. Now we are updating the discriminator loss or

[09:47] discriminator parameters. If discriminator parameters are updated, generator has no role. ϕ will be uh you know kept constant. So this is what is this entire leftsided term. So our aim is to minimize the loss with respect to the parameters θ . And what is that? You have to minimize the loss function LD. And what is LD loss at a discriminator side? What are these two? These are the parameters. What are parameters we have in this entire loss function? We have a θ as well as ϕ . But since we are dealing with a discriminator this generator part will remain um fixed or uh you know freeze froz you will you will not be dealing anything with respect to the ϕ . Now uh what is this LD? LD is nothing but

[10:41] negative of this entire. So that is what has been written here. This minimum with respect to θ is retained as it is. this whole LD this entire is being substituted by this particular formula minus of the loss. So this is what is the discriminator loss function. Now uh if you uh recall once again when we now whatever we have we have in terms of minimization minimum of. So can you recall the concept minimum um if you want to minimize something it is also equivalent to maximization of it. Right?

[11:22] So if minimum of minus something is equivalent to the maximization of it right do you agree to this? So minimization minus AD is equivalent to maximization positive one. Okay. So now whatever we have obtained in terms of negative in the previous slide. Can I write that in terms of positive? So look at this. This is what we have obtained from our previous where this is LD. So minimization LD is equivalent to maximization. So can we write in this way the same one instead of writing minimum minus can I write a maximum here by using this particular concept. So if a minimum is LD maximization can I denote it as capital V D right u V is a value function and D is at a discriminator side what did we do since

[12:22] there is a math that we are entering I'll be slow and then I'll be repeating here again from our previous one we had calculated discriminator loss for a real and discriminator loss for fake or generated then the total loss is given by LD is $L_{\text{real}} + L_{\text{fake}}$. Now you have substituted the formula um value and then you have obtained there is an minus in both you have taken minus out by taking a minus you have obtained this particular formula we had obtained up till this what is our ultimate aim of a neural network aim is to minimize the loss therefore minimization with respect to the parameters θ and the θ here are weights and bias if it is an simple neural network or it can be the kernel weights if it is a CNN based.

[13:15] Okay. Now, minimization is equivalent to maximization. Therefore, whatever if minimization is denoted with the term LD, max since you have now converted into maximization, we are writing it as WD. That is where we have written here minimizing LD is equivalent to maximizing V value function and D is at a discriminator. I hope up till this point it is clear to us. Right now the next step is whatever the loss function that we have written here this particular loss function is taking we uh this is only for one real sample and one generated sample. So this is only for one uh sample but when uh so this is what has been written here. There is one real image X and one noise sample zed.

[14:10] But when you during the training uh phase it is not one single image that we will be passing to the model. We will pass an mini batch of samples. Right? So during a training we uh usually don't take one. we uh we use many real and generated samples. If that is the case, the loss function cannot be just uh this simple. How is that you'll be doing? Is it not for every of the sample you have to calculate this and for many samples overall you have to take an average. So is it not uh this is what we are doing for a real sample we are now writing this term as expectation. What exactly is expectation here? Now what what is this whole term meaning we are taking a many such x from where from an real data

[15:05] distribution p data of x for each sample we are computing this and then we are taking the average of all of it. So is it not we are doing this? Take a many samples X from a real data distribution P data of X. Then compute uh this particular $\log d \theta$ of X for each sample and then take the average. Then you can ask the question why is that we have written in this way. Can we also write this like this? Anyhow you are adding all of this and then you are taking the average. So which is nothing but one by that is you have taken you are performing the summation of each of these terms and how many ever samples you have taken you are taking the average of it. So either you write in terms of expectation or you can write in

[15:55] terms of average one summation and average the this term is equivalent to this term. So this is with respect to real what we have considered is this. But for a discriminator we are also passing an generated samples right. So for generated samples how is that? This is the term for generated one. And here you are not considering one. You have said that there are many such generated samples. If there are many such generated samples for each sample you compute this and then you will take the average of it. Right? So that is equivalent to this term. So in this particular slide what is that we have understood uh the minimization is equivalent to maximization. If minimization is denoted as LD the maximization

[16:51] we will be denoting it as the value function for at a discriminator site V D. Now uh this is with respect to a single sample. But when you take an uh training you will have many such a real samples many such um you know generated samples. So for many real samples you will be taking the average of you will compute for each

term and you will take the average of it. So we call it as an expectation. Same thing you are doing it for generated samples as well. So once this is done now let us write what is the total value function or an objective function of an uh discriminator. So this is the final discriminator objective function. So if [snorts] you see this you remember we have LD is for minimization but later we have converted

[17:45] it into a maximization. Therefore can we write it as V and D ? So it is v_d value function of a discriminator d . Now this term indicates here you have two θ as well as five. So this semicolon in this particular term indicates only θ is being optimized while this five is kept fixed. Right? So this is what we have understood during uh training uh and during a back propagation at only at a discriminator site discriminator gets updated. During uh the training of a generator the generator gets updated. Now what we are discussing here is the discriminator. So because it is a discriminator the generator has no role um to get updated here. So this particular parameters gets fixed right. So that is what is the left

[18:42] hand term meaning. Now what exactly is the right side the what is this part? So here uh earlier we had seen for if you see if you go back here we had seen for a single x and a single z but what we are discussing here in the next slide here is for many such a real samples and many such generated samples. So many real and many generated is it not the expectation? So therefore we are writing in this form. I hope uh this is clear to all of us.

[19:20] Okay. Now once this is done what is our ultimate aim? Aim here is the discriminator is maximizing this objective function or maximizing this value function. So now so what exactly it is maximizing this entire term. So rather than write rewriting once again the whole equation here I'm showing a maximizing this term right maximization with respect to parameters θ . Now you can ask the question why discriminator for a discriminator we have used the term maximization here. Is it because from negative you have converted into positive or minimization to maximization. Therefore you you have got this maximization here. So let us see why a discriminator always aims to maximize this that is the point of discussion now. So whatever we

[20:21] have seen uh in theory in the previous session that will be made more clear here mathematically. Okay. Uh what is the point of discussion? Understanding why discriminator maximizes this term. What a term this entire. Now look at this uh in this particular equation. Recall a discriminator is the one where you are passing real samples as well as generated samples. Now uh whatever the equation that we have written here is one is for the real this is for the real and this is for the generated. So both have been passed to an discriminator.

[20:57] Now uh uh once again recall for an discriminator when a real samples have been passed what is the aim of a discriminator? It has to give the probability for a real samples closer to one. Now let us understand why it gives closer to one. That

part also will be made clear here. So what we are discussing is the first term that is the real sample term. for a real sample term. Is it not? We have to consider only so much because this is the first term is something which is dealing with the real images. Why real images?

[21:34] Because you are passing X . X is the real samples. So for a real sample we are considering the $\log d_{\theta}$ of X term. In this what is d_{θ} of X ? Is it not the discriminator's probability predicted probability that the input X is real? Okay. Now when you take a discriminator and when you pass the real images, what should be the uh probability what should be the output? The output should be closer to one. So that is where d_{θ} of x value uh the aim here is the desired output is the value d_{θ} of x that is the predicted probability value should be closer to one. This is what is the aim here. Okay.

[22:20] Now at the beginning you have still not started any training. So initially you have randomly assigned some values. So the θ value whatever we have the θ parameters weights and bias are all randomly assigned. So because of randomly assigned now a real image is being passed to a discriminator. So what will be the predicted probability? Do you think the for the first time when real images have been passed discriminator will give the very high probability? No. Because you have still not trained the model, how will the model knows which is real, which is generated, right? So when samples have been passed, real samples have been passed and at the beginning of the training, not necessarily that the

[23:02] discriminator has given a very high probability, not necessary. Let us assume discriminator has told has given this particular probability 0.1 which means the discriminator is performing very poor here. Even for a real image the probability value is very very small which means only 10% it is saying that it is real. No it's not correct right. So and why this is happening it is at the beginning of the training where you have still not started a training and the parameters are randomly assigned. So because of this d_{θ} of x the predicted probability value is very poor and here it is 0.1 for an example. Okay.

[23:44] If this is 0.1, what is our um function here? It is $\log d_{\theta}$ of x . If d_{θ} of x is 0.1, this $\log 0.1$, what is the value? We are getting very low negative value. It is very low value here. Right? So now, so with this what are we understanding? Is it not just with this value minus 2303 is it not we are understanding that discriminator performance on real images is very poor right so now that is been passed as an feedback to an discriminator gradients in the form of a gradients now over a period of time is it not uh that discriminator gets trained so after some training say for example the discrim discriminator has given this particular probability 0.5.

[24:40] Now when the discriminator has given this particular probability d_{θ} of x now what exactly is $\log 0.5$ the $\log 0.5$ is this is it not we are minimizing the

value uh earlier from this value to this value you see is it not we are um you know uh improving uh can we say in this way now uh next you see again over a period of time uh um the discriminator has further improved. So discriminator value is now 0.9. For example, after a certain iterations of training, now discriminator has for a real images discriminator has given this particular probability.

[25:26] Now uh uh after you obtained 0.9 again you calculate the loss function. What is loss here? $\log 0.9$ is it not? We have obtained this value right which is now closer to zero. So uh is it not now you are increasing the value ultimately aim is to maximizing what is that you have to maximize you have to maximize this term. So is it not from here you have improved from here you have improved right. So now you are maximizing this value. Now uh um go back and then uh uh understand this. What is that we have written? We have written $\log \theta$ of x which is written here. Now what is that you should substitute here so that we get the maximum value. So what is the value that you need to substitute here?

[26:22] It is the probability right? Probability is at the most it can give the value between zero and one. So can I have if this entire term is equivalent to one then $\log$ one is zero right? So uh did you get this concept? This is important. Uh whatever we have seen in the previous theory that will be made clear here. We are saying that discriminator should maximize this term. If discriminator has to maximize this term for what the discriminator has to maximize this term here for what value of this discriminator you know this $\log$ term will be maximum. So since it is an probability for a real image x when you pass either it can predict the value zero to one only right. So if you want this $\log$ term to be maximum which means

[27:18] this $\log \theta$ of x should be one. This is a maximum value that you can give. I hope this concept is clear to you. That is the reason why you know um $\log \theta$ of x um you know model going to give this as one because ultimately it is not exactly one. is closer to one maximization because as $\log \theta$ of x increases $\log \log \theta$ of x also is getting increased you see from 0.1 uh to 0.5 to 0.9. So $\log \theta$ of x is getting increased as uh uh this value is getting increased. Look at this minus uh 2.3 to minus 0.6 6 to minus 0.1.

[28:05] So as the $\log \theta$ of x increases $\log$ value is also increasing. But what is the maximum limiting value? Where $\log \theta$ of x can be only one. At the most it can be one. If this is one you are reaching to a maximum value $\log$ one is zero. Therefore maximizing maximizing the real image term pushes $\log \theta$ of x value closer to one. So I hope this is clear to you. Now let us see what exactly happens for an generated samples. So for generated samples which means we are now dealing with this entire concept this entire term. Now what exactly it is saying you have to maximize this term. If you have to maximize this term, what term? $\log$ of that entire. If $\log$ of that entire has to be maximized, what should be the best

[29:04] possible value of this? So that that entire log term will be maximized. That is what we are discussing. So if you recall we are since we are dealing with the generated samples we are considering this term. In this what exactly is G_5 of Z ? This is a generated sample. That generated sample is then passed to a discriminator. What is that discriminator will give? Discriminator will give the predicted probability for the generated samples. Okay. Uh now uh recall it is a discriminator. For a discriminator you are passing an generated samples. Therefore it's probability value should be closer to zero. This we know it very very well.

[29:50] Now if this value is closer to zero which means we are now saying that d_{θ} this entire term value should be closer to zero then you're going to maximize this term is what we are trying to understand from this. Now let us see why in which scenario we are going to maximize this at the beginning of the training definitely discriminator is not very well uh trained here. So um initially discriminator may incorrectly classify it which means when you have passed and generated samples ideally it has to give the probability closer to zero but not necessarily right. So because weights are randomly assume assigned. Let us assume that for a generated sample discriminator has given the value 0.9.

[30:43] Right? If this is the predicted value what is there in the formula? It is 1 minus because it is 1 minus. So look at this 1 minus 0.9 is a smaller value that is 0.1. Now that smaller value you pass it to the log $\log 0.1$ is it not you are getting an um very low negative value right you have obtained very low negative value here. So which means discriminator performance on the generated samples is very poor right. So what is that we have to do now? So next is as the training progresses after some training let us now assume the discriminator has produced this particular value right for what for a generated sample discriminator has given this particular value. Look at this 1 minus this you have obtained 0.5 $\log 0.5$ you have

[31:42] obtained this okay after further training so let us assume now discriminator has given this particular probability right 0.1 which means discriminator probability is now closer to zero right if it is closer to zero it means it's very good when this is closer to zero 0.1 $1 - 0.1$ is 0.9 $\log 0.9$ Look at this value. Are you getting at what we are discussing? Ultimately our aim is to maximize this term. If you have to maximize this term, what should be the value? The maximum possible value this should be zero. Then $1 - 0$ is 1. $\log 1$ is 0. Right? That is the maximum value.

[32:34] Which means this term should have a value closer to zero. Is it not? We are seeing all of this here. Uh if you look into the previous one, uh here initially the value what we have obtained is 0.9. It is very high value. From there you have obtained this value. Again when you further go when you look into this here you have a value 0.5. Then you have obtained this value. Further you have got 0.1 which means you are now moving closer to zero. When you are moving closer to zero look at this term right this is also um this value of what exactly is happening here? What is the relationship

between 0.1 as you are decreasing the value? What exactly is happening to log value? Is it not log value is increasing? You first had minus

[33:33] 2.3 to minus0.6 now it is minus0.1 is it not now moving to zero. So is it not as um the predicted value decreases closer to zero log value is increasing. So you see here finally we have to maximize this whole term. If you have to maximize this term for what value you are going to get the maximum value. So this value should be closer to 0. Then only 1 minus 0 is 1. Log 1 is 0. Right? Uh I hope this is clear to you. So that is the reason why the discriminator pushes this particular term closer to zero. Right? What is the relation? As I said, as this value decreases, one minus this uh this term increases and the log term increases. What is our ultimate aim? Aim is to minim maximize this term.

[34:35] What term the discriminator is the aim is to maximization. So if it has to maximize, we have understood already how this is maximizing. Now we are understanding how this term has to be maximized. So if your aim is to maximize log of this entire term that is possible only when uh this value becomes closer to zero right so that is the reason why discriminator maximizes the value function. I hope now it is clear to you.

[35:09] From theory we have seen that for a real image the predicted probability should be closer to one. For a generated images the predicted probability should be closer to zero. So that is what has been made clear from this. Now so with this we have now come to an generator objective function. What should be the generator objective function? You see how it has been written because it is a value function for a generator. We have written it as VG and it is a generator.

[35:40] Now your aim is to improve the generator one. So therefore generator parameters are optimized. Theta that is discriminator parameters are fixed and we will have only one objective function here. Overall this is for the real images and this is for an generated images. So for the same now let us understand what exactly is uh generator objective function. So um in this entire if you look into the first term the first term that is this term it is all with respect to real images and then the discriminator.

[36:20] So um when you're dealing with the generator part we don't have any work with respect to first term right look at this the first term contains only the real samples and then the discriminator part therefore it does not depends on generator parameters five therefore generator cannot change anything this with respect to this first term. Now let us see when uh you are training the generator what exactly comes into the play. It is only the second term. So second term has the generated samples.

[36:56] Therefore during the generator update we optimize only the second term. We optimize only this term. Okay. Now recall from our theory concept you are now dealing with generator. Your aim is to improve the generator. Now generator has generated the samples. This generated samples are given to a discriminator. What

is the role of a discriminator? Now here for a generated samples we are discussing at a generator part. So when [snorts] generate generator has generated the samples when it is given to a discriminator discriminator should give the high probability right. Why it is why it has to give the high probability? when it gives the high probability then uh and its ground truth is one there. Therefore

[37:47] when you uh compute the gradients and when you give that feedback back to the generator generator now understands that okay whatever images I'm generating those images are realistic images correct so which means for a generated samples the aim is discriminator should give the high probability so let's recall whatever we have seen the generator wants the discriminator to classify the generated samples is real which means the probability of this should be closer to one. Right? Now if this value is closer to one, what exactly is the problem? Go back. This value should be closer to one is what we are saying. If this value is closer to one or one minus one becomes zero. So is it not this term becomes zero. If this

[38:39] value is closer to one, $1 - 1$ value is closer to zero. If this is zero, what happens to log term? $\log 0$. $\log 0$ is undefined. So this entire term has become now zero. $\log 0$ is undefined or it is minus infinity. [snorts] Then what is the aim? What is the aim of a uh the aim of a discriminator was to maximize? What is the aim of a generator? Now is it not it has to minimize this particular term $\log$ this entire term has to be minimized. So therefore you see here so the generator tries to make this term as small as possible. Uh this particular $\log$ of this entire term as small as possible. Therefore, the aim of an generator here is to minimize this entire term. Okay. So, um if you have to visualize, you can see this. Um

[39:41] ultimately the aim here is this value should be closer to one. So, as this value gets closer to one, look at this. What exactly is happening? the value is getting minimized right minus 6 is you know the value is very small here so this term is small so therefore the generator objective function here is to minimize so I hope this is clear to you although it is very complicated um concept but it is you know as you understand you will be able to get this concept clearly okay with Thus uh let us see what is the GAN overall GAN objective function. What we have seen in the beginning of the session is discriminator objective function aim here is to maximize right the discriminator objective function is to

[40:39] maximize with respect to θ and just some time back in the previous slide we have seen what is the aim of an generator. The aim of a generator is to minimize. So the aim of a generator here is to minimize with respect to ϕ . But when we take a GAN here we don't have independently generator independently discriminator we have generator discriminator entire together as one. So combining both um gives a original GAN min max objective. How is that you will combine? One is minimum second is maximum. Right? The aim of a generator is to minimize generator here is to minimize and the aim of a discriminator is to maximize. So you are combining both

of these by combining both we are obtaining it as min max. So can we now

[41:33] write this as min max that entire v is whatever we have seen so far right this is this entire is the v . Now here v is with respect to g uh θ is with respect to d . So can we write in terms of g and d . So therefore one is minimizing and one is maximizing. So therefore the goals here are opposite. What are the go? What what is opposite here? Now discriminator wants the generated uh uh samples to give the probability closer to zero but generator wants the same generated samples to receive the value closer to one. So is it not um there are two opposite um you know objectives here for for a generator wants whatever the samples generated samples I'm receiving I should give the probability closer to

[42:38] zero but what is generator expectation generator is saying that whatever images I am generating generator is generating when we give it to a discriminator discriminator should give the high probability so one is minimizing the objective function the second is maximizing the objective function. Therefore, go back and then understand we have a a he is adversarial. What is adversarial here? Is it not two or competing with each other? One is minimizing, second is maximizing. So this is all about the GAN um you know uh the architecture diagram and then understanding the same architecture mathematically. So this is the objective function. With this we have come to an end of this particular session. In the

[43:25] next session we are going to see a very small topic called GAN convergence and equilibrium. Thank you.